

谐振子递推公式推导

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本文根据量子力学学习中的各种资料整理所得，如有错误，请反馈

一维线性谐振子

哈密顿量：

$$\hat{H} = -\frac{\hbar^2}{2\mu} \frac{d^2}{dx^2} + V(x)$$

$$V(x) = \frac{1}{2} \mu \omega x^2$$

波函数：

$$\psi_n(x) = N_n e^{-\alpha^2 x^2 / 2} H_n(\alpha x)$$

能量：

$$E_n = \left(\frac{1}{2} + n \right) \hbar \omega$$

其中 $n = 0, 1, 2, \dots$

$$\alpha = \sqrt{\frac{\mu \omega}{\hbar}}, \quad N_n = \sqrt{\frac{\alpha}{n! 2^n \sqrt{\pi}}}$$

则能量本征函数, 当参数方程条件满足时, 波函数的的一个解是Hermite多项式 $H_n(\xi)$ (满足物理边界条件), 另一个解为无穷级数 (在物理上不满足边界条件)

前几个 *Hermite* 多项式为:

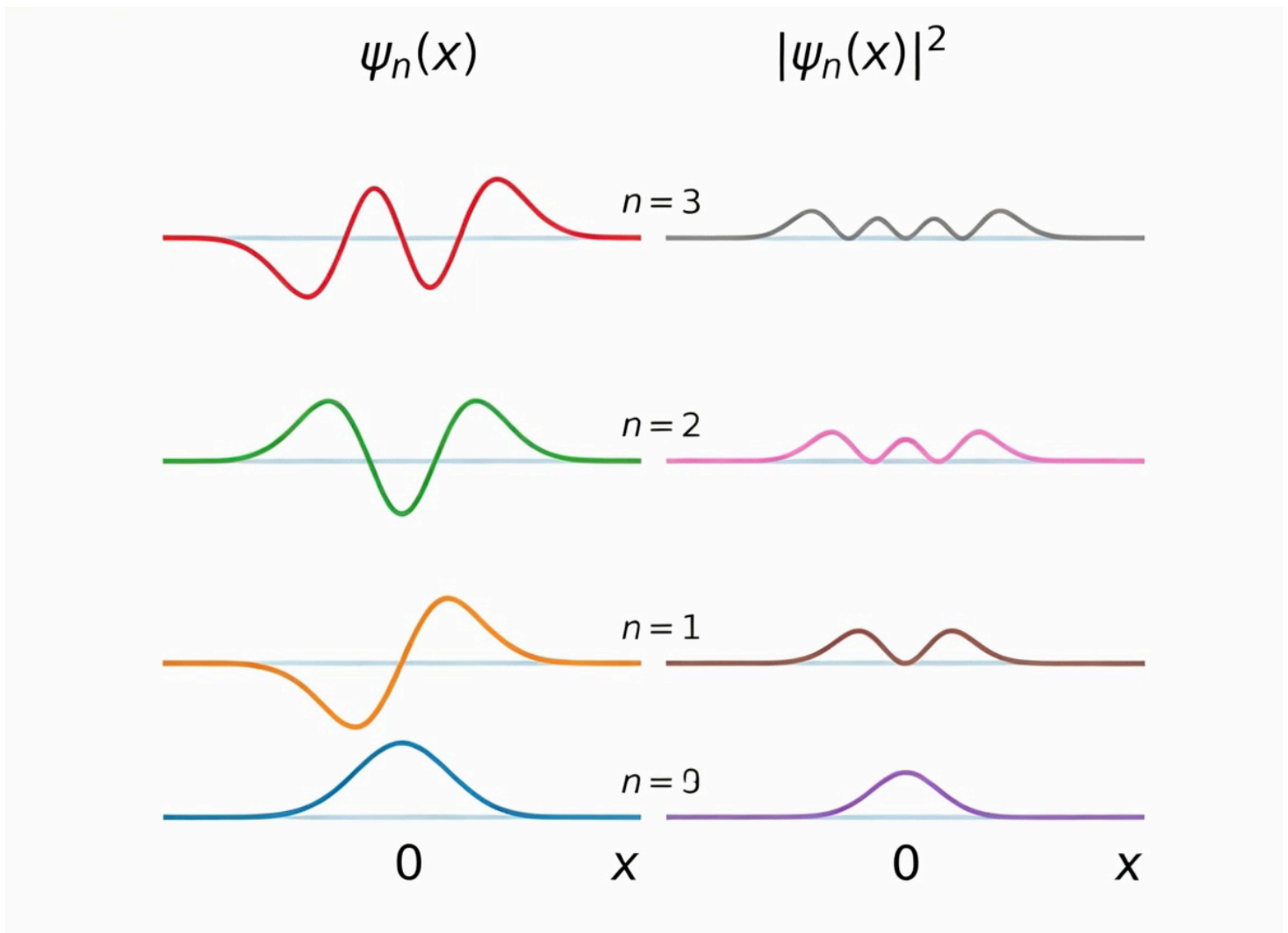
$$\begin{aligned}
 H_0(\xi) &= 1 \\
 H_1(\xi) &= 2\xi \\
 H_2(\xi) &= 4\xi^2 - 2 \\
 H_3(\xi) &= 8\xi^3 - 12\xi \\
 &\dots\dots\dots \\
 H_n(x) &= (-1)^n e^{x^2} \frac{d^n}{dx^n} (e^{-x^2})
 \end{aligned}$$

上式最后一项被称为Hermite多项式的罗德里德公式。

则前几个能级上的谐振子能量本征函数为：

$$\begin{aligned}
 \psi_0(x) &= \frac{\sqrt{\alpha}}{\pi^{1/4}} e^{-\frac{1}{2}\alpha^2 x^2} \\
 \psi_1(x) &= \frac{\sqrt{2\alpha^3}}{\pi^{1/4}} x e^{-\frac{1}{2}\alpha^2 x^2} \\
 \psi_2(x) &= \frac{1}{\pi^{1/4}} \frac{\alpha}{2} (2\alpha^2 x^2 - 1) e^{-\frac{1}{2}\alpha^2 x^2} \\
 \psi_3(x) &= \frac{\sqrt{3\alpha^3}}{\pi^{1/4}} x (\frac{2}{3}\alpha^2 x^2 - 1) e^{-\frac{1}{2}\alpha^2 x^2}
 \end{aligned}$$

直接求解很复杂，虽然图像看起来很简单：



阶梯算符法：

一维谐振子降算符 $\hat{a}$ 与升算符 $\hat{a}^\dagger$ 的定义是：

$$\hat{a} = \sqrt{\frac{\mu\omega}{2\hbar}} \left(x + \frac{i}{\mu\omega} \hat{p} \right)$$

$$\hat{a}^\dagger = \sqrt{\frac{\mu\omega}{2\hbar}} \left(x - \frac{i}{\mu\omega} \hat{p} \right)$$

其中 $\hat{p}$ 是 x 方向的动量算符， $\hat{a}$ 与 $\hat{a}^\dagger$ 满足对易关系：

$$[\hat{a}, \hat{a}^\dagger] = 1$$

由 $\hat{a}$ 和 $\hat{a}^\dagger$ 可以构成厄米算符 $\hat{N} = \hat{a}^\dagger \hat{a}$ 。 $\hat{N}$ 的本征值为 $n=0, 1, 2, \dots$ 。令 $\hat{N}$ 的本征态为 $|n\rangle$

$$\hat{a}^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle$$

$$\hat{a} |n\rangle = \sqrt{n} |n-1\rangle$$

一维谐振子哈密顿量 $\hat{H} = \frac{\hat{p}^2}{2\mu} + \frac{1}{2}\mu\omega^2 x^2$ 可以通过 $\hat{N}$ 表示为:

$$\hat{H} = (\hat{N} + \frac{1}{2})\hbar\omega$$

则 $\hat{H}$ 的本征值为 $E_n = (n + \frac{1}{2})\hbar\omega, n = 0, 1, 2, \dots$

由 $\hat{a}$ 与 $\hat{a}^\dagger$ 计算可得:

$$\hat{x} = \sqrt{\frac{\hbar}{2\mu\omega}}(\hat{a}^\dagger + \hat{a})$$

$$\hat{p} = \sqrt{\frac{\mu\omega\hbar}{2}}i(\hat{a}^\dagger - \hat{a})$$

作用于波函数:

$$\begin{aligned}\hat{x}|n\rangle &= \sqrt{\frac{\hbar}{2\mu\omega}}(\hat{a}^\dagger + \hat{a})|n\rangle \\ &= \sqrt{\frac{\hbar}{\mu\omega}}\frac{1}{\sqrt{2}}(\hat{a}^\dagger|n\rangle + \hat{a}|n\rangle) \\ &= \sqrt{\frac{\hbar}{\mu\omega}}\frac{1}{\sqrt{2}}(\sqrt{n+1}|n+1\rangle + \sqrt{n}|n-1\rangle) \\ &= \frac{1}{\alpha}(\sqrt{\frac{n+1}{2}}|n+1\rangle + \sqrt{\frac{n}{2}}|n-1\rangle)\end{aligned}$$

$$\text{又}\hat{p} = \frac{\hbar}{i}\frac{d}{dx}$$

$$\begin{aligned}\frac{d}{dx}|n\rangle &= \frac{i}{\hbar}\sqrt{\frac{\mu\omega\hbar}{2}}i(\hat{a}^\dagger - \hat{a})|n\rangle \\ &= -\sqrt{\frac{\mu\omega}{\hbar}}\sqrt{\frac{1}{2}}(\hat{a}^\dagger|n\rangle - \hat{a}|n\rangle) \\ &= \sqrt{\frac{\mu\omega}{\hbar}}(\sqrt{\frac{n}{2}}|n-1\rangle - \sqrt{\frac{n+1}{2}}|n+1\rangle) \\ &= \alpha(\sqrt{\frac{n}{2}}|n-1\rangle - \sqrt{\frac{n+1}{2}}|n+1\rangle)\end{aligned}$$

由上式可以继续推导。

$$\because \hat{x}|n\rangle = \sqrt{\frac{\hbar}{\mu\omega}}(\sqrt{\frac{n+1}{2}}|n+1\rangle + \sqrt{\frac{n}{2}}|n-1\rangle)$$

$$\begin{aligned}\therefore \hat{x}^2|n\rangle &= \sqrt{\frac{\hbar}{2\mu\omega}}\sqrt{\frac{\hbar}{\mu\omega}}(\hat{a}^\dagger + \hat{a})(\sqrt{\frac{n+1}{2}}|n+1\rangle + \sqrt{\frac{n}{2}}|n-1\rangle) \\ &= \frac{1}{\sqrt{2}}\frac{\hbar}{\mu\omega}(\sqrt{\frac{n+1}{2}}\hat{a}^\dagger|n+1\rangle + \sqrt{\frac{n}{2}}\hat{a}^\dagger|n-1\rangle + \sqrt{\frac{n+1}{2}}\hat{a}|n+1\rangle + \sqrt{\frac{n}{2}}\hat{a}|n-1\rangle) \\ &= \frac{\hbar}{\mu\omega}(\frac{\sqrt{n+1}}{2}\sqrt{n+2}|n+2\rangle + \frac{\sqrt{n}}{2}\sqrt{n}|n\rangle + \frac{\sqrt{n+1}}{2}\sqrt{n+1}|n\rangle + \frac{\sqrt{n}}{2}\sqrt{n-1}|n-2\rangle) \\ &= \frac{1}{\alpha^2}(\frac{\sqrt{(n+1)(n+2)}}{2}|n+2\rangle + (n+\frac{1}{2})|n\rangle + \frac{\sqrt{n(n-1)}}{2}|n-2\rangle) \\ &= \frac{1}{\alpha^2}(\frac{1}{2}\sqrt{n(n-1)}|n-2\rangle + (n+\frac{1}{2})|n\rangle + \frac{1}{2}\sqrt{(n+2)(n+1)}|n+2\rangle)\end{aligned}$$

$$\because \frac{d}{dx}|n\rangle = \sqrt{\frac{\mu\omega}{\hbar}}(\sqrt{\frac{n}{2}}|n-1\rangle - \sqrt{\frac{n+1}{2}}|n+1\rangle)$$

$$\begin{aligned}\therefore \frac{d^2}{dx^2}|n\rangle &= \frac{i}{\hbar}\sqrt{\frac{\mu\omega\hbar}{2}}i(\hat{a}^\dagger - \hat{a})\sqrt{\frac{\mu\omega}{\hbar}}(\sqrt{\frac{n}{2}}|n-1\rangle - \sqrt{\frac{n+1}{2}}|n+1\rangle) \\ &= \frac{i}{\hbar}\frac{1}{\sqrt{2}}i\mu\omega(\sqrt{\frac{n}{2}}\hat{a}^\dagger|n-1\rangle - \sqrt{\frac{n+1}{2}}\hat{a}^\dagger|n+1\rangle - \sqrt{\frac{n}{2}}\hat{a}|n-1\rangle + \sqrt{\frac{n+1}{2}}\hat{a}|n+1\rangle) \\ &= -\frac{\mu\omega}{\hbar}(\frac{\sqrt{n}}{2}\sqrt{n}|n\rangle - \frac{\sqrt{n+1}}{2}\sqrt{n+2}|n+2\rangle - \frac{\sqrt{n}}{2}\sqrt{n-1}|n-2\rangle + \frac{\sqrt{n+1}}{2}\sqrt{n+1}|n\rangle) \\ &= \alpha^2(-\frac{n}{2}|n\rangle + \frac{\sqrt{(n+2)(n+1)}}{2}|n+2\rangle + \frac{\sqrt{n(n-1)}}{2}|n-2\rangle - \frac{n+1}{2}|n\rangle) \\ &= \alpha^2(\frac{1}{2}\sqrt{n(n-1)}|n-2\rangle - (n+\frac{1}{2})|n\rangle + \frac{1}{2}\sqrt{(n+2)(n+1)}|n+2\rangle)\end{aligned}$$

则有：

$$\begin{aligned}\hat{x}^2|n\rangle &= \frac{1}{\alpha^2}(\frac{1}{2}\sqrt{n(n-1)}|n-2\rangle + (n+\frac{1}{2})|n\rangle + \frac{1}{2}\sqrt{(n+2)(n+1)}|n+2\rangle) \\ \frac{d^2}{dx^2}|n\rangle &= \alpha^2(\frac{1}{2}\sqrt{n(n-1)}|n-2\rangle - (n+\frac{1}{2})|n\rangle + \frac{1}{2}\sqrt{(n+2)(n+1)}|n+2\rangle)\end{aligned}$$